

COVERAGE_LEDGER

HelixCode CONST-048 Coverage Ledger — Submodule × Feature × Platform × Invariant

Schema: `SCHEMA_VERSION = 1.0.0` (see `SCHEMA.md`) **Last regenerated:** 2026-05-18 (round 68 initial publication) **Source roster:** `docs/improvements/submodule_owned.txt` (68 owned submodules) **Companion ledger:** `docs/coverage/ledger.md` (feature × invariant rollup, round 41+) **Generator:** `scripts/generate-coverage-ledger.sh` **Mandate:** CONST-048 (Full-Automation-Coverage) / constitution §11.4.25; CONST-035 anti-bluff posture; Article XI §11.9

How to read this ledger

Each row = (Submodule, Feature, Platform). Six invariant cells per row (I1..I6) hold one of: PASS, PENDING_FORENSICS:, OPERATOR-BLOCKED:, UNCONFIRMED:, SKIP-OK: #<ticket>, PARTIAL, N/A. Overall column is mechanical rollup. Notes column carries the evidence reference for any PASS marker.

Round 68 deliverable = **structure + honest conservative initial population**. Default cell is UNCONFIRMED:. PASS markers only where CONTINUATION.md round narrative documents captured evidence. Subsequent rounds promote UNCONFIRMED: → PASS by adding the round reference in Notes.

Auto-marking everything PASS would itself be a CONST-035 PASS-bluff at the governance layer — explicitly forbidden by the round 68 brief.

CONST-048 invariants

#	Invariant
I1	Anti-bluff posture with captured runtime evidence (CONST-035 / Article XI §11.9)
I2	Proof of working capability end-to-end on target topology (CONST-050(A) — no mocks beyond unit tests)
I3	Implementation matches the documented promise (§11.4.12)
I4	No open issues/bugs surfaced (Issues.md cross-check + source-tree TODO(round-N) / BUG #NN: grep)
I5	Full documentation in sync (§11.4.12)
I6	Four-layer test floor (pre-build + post-build + runtime + paired mutation)

Coverage rollup (round 68 baseline)

Metric	Count
Owned submodules (roster)	68
Rows in ledger	68 (one per submodule, whole-module feature row each)
PASS Overall	0
PARTIAL Overall	8
PENDING_FORENSICS Overall	0
OPERATOR-BLOCKED Overall	0
UNCONFIRMED Overall	60

Conservative posture per CONST-035: no full-row PASS until every invariant for every platform has captured evidence. PARTIAL rows hold the documented-PASS work from CONTINUATION rounds 23/37/41/53/60/62/63/64/65 (the round-specific evidence appears in the cell + Notes).

Ledger rows

Submodules with documented PASS-on-some-invariants (PARTIAL Overall)

Submodule	Feature	Platform	I1 anti-bluff	I2 e2e-working
submodules/ auto_temp	grid-search temperature tuning	linux,macos,windows	PASS (round 23 fix 127be62)	PASS (round 37 wired backends)
submodules/ storage	recording S3 sync	linux,macos	PASS (round 37 acecf73)	PARTIAL (round 37 en SKIP-OK on real S3)
submodules/ llm_orchestrator	OpenCode CLI adapter	linux,macos	PASS (round 64 a9493fa)	PASS (round 64 real op v1.14.41 invocation)
submodules/ llm_orchestrator	SimpleAgentPool + 5 builder factories	all-platforms	PASS (round 50+ per CONTINUATION)	UNCONFIRMED:
submodules/ helix_specifier	real DebateFunc	all-platforms	PASS (round 65 f2cb17a)	PASS (round 65 LLMR interface + LLMBackedDebateFun
submodules/ llm_provider	provider error sentinel coverage (17 providers)	all-platforms	PASS (round 63 5171edf)	PASS (round 63 full llm 52.868s)

Submodule	Feature	Platform	I1 anti-bluff	I2 e2e-working
submodules/ llms_verifier	provider/model verification (CONST-036)	all-platforms	PASS (CONST-036/037 wiring per ledger.md)	PARTIAL (groq+openrouter+mistral) live-verified per ledger.
submodules/ models	model metadata source	all-platforms	PASS (per CONST-036 ledger)	PARTIAL (per LLMsV2 integration above)

All other owned submodules — UNCONFIRMED baseline (round 68)

Each of the rows below is a single whole-module placeholder. Future rounds split into per-feature rows when audit work captures evidence. Until then, the rows hold UNCONFIRMED across every invariant — which is the honest position per CONST-035 (“never claim PASS without captured evidence”).

Submodule	Feature	Platform	I1 anti-bluff	I2 e2e-working	I3 doc-mat
challenges	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
containers	whole-module	linux,containers	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/ doc_processor	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/ helix_llm	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/ helix_memory	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/ vision_engine	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/agentic	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/auth	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/ background_tasks	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/ benchmark	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/cache	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:

Submodule	Feature	Platform	I1 anti-bluff	I2 e2e-working	I3 doc-mat
submodules/claritas	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/concurrency	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/config	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/conversation	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/database	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/doc_processor	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/document	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/embeddings	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/event_bus	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/filesystem	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/formatters	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/gandalf_solutions	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/hyper_tune	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/il8n	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/i_llm	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/lazy	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/leak_hub	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/llm_ops	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/llm_orchestrator	whole-module (vasic-	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:

Submodule	Feature	Platform	I1 anti-bluff	I2 e2e-working	I3 doc-mat
	digital fork)				
submodules/llm_provider	whole-module (vasic-digital fork)	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/mcp_module	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/memory	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/messaging	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/middleware	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/models	whole-module (vasic-digital fork)	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/normalize	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/observability	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/optimization	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/ouroborous	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/planning	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/plinius_common	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/plugins	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/rag	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/rate_limiter	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:

Submodule	Feature	Platform	I1 anti-bluff	I2 e2e-working	I3 doc-mat
submodules/recovery	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/red_team	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/self_improve	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/skill_registry	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/streaming	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/tool_schema	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/toon	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/vector_db	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/veritas	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/vision_engine	whole-module (vasic-digital fork)	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
submodules/watcher	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
github_pages_website	whole-module	headless	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
helix_agent	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
helix_qa	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
panoptic	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:
security	whole-module	all-platforms	UNCONFIRMED:	UNCONFIRMED:	UNCONFIRMED:

Cross-reference points

- **CONTINUATION.md** — canonical source of round-by-round PASS justifications. Cite round NN in Notes for any PASS marker.

- **ledger.md** — feature × invariant rollup (F01..F30 catalogue). Cross-check that PASS rows here align with F-cell PASS in ledger.md.
- `docs/improvements/submodule_owned.txt` — canonical owned-submodule roster; every entry **MUST** have ≥1 row here.
- `scripts/generate-coverage-ledger.sh --check` — enforces submodule coverage + schema invariants; CI gate.
- **Issues.md** (per-submodule, future) — feeds I4 once introduced per CONST-057 / §11.4.16.

How PASS markers get promoted (round N+ workflow)

1. Round agent completes work on a submodule.
2. Round agent captures evidence (test logs, integration runs, real-binary invocations).
3. Round agent updates CONTINUATION.md with round narrative + commit SHA.
4. Round agent (or operator) edits this ledger:
 - Find the row for (submodule, feature, platform) — or split a whole-module row into per-feature rows.
 - Promote UNCONFIRMED: → PASS for the specific invariant the work addresses.
 - Add to Notes: round NN <SHA> + brief evidence pointer.
5. Commit ledger update in the same commit as the work (per CONST-044 CONTINUATION-sync discipline).
6. `bash scripts/generate-coverage-ledger.sh --check` **MUST** pass.

Anti-bluff guarantees (round 68)

1. Default cell = UNCONFIRMED: — never PASS. (60 of 68 rows are UNCONFIRMED Overall at round 68 publication — honest baseline.)
2. PASS marker requires either CONTINUATION round reference or pasted-evidence path in Notes (enforced by `--check` mode of generator).
3. Every owned submodule has ≥1 row (enforced by `--check`).
4. Status vocabulary is closed (enforced by `--check`).
5. Schema version bump on column changes (enforced by `SCHEMA.md` discipline).
6. PASS-by-default is forbidden — it would be a CONST-035 PASS-bluff at the governance layer, equivalent severity to a false-success test result per Article XI §11.9.

Operator mandate (verbatim, 2026-05-19, preserved per CONST-049 §11.4.17)

“all existing tests and Challenges do work in anti-bluff manner - they **MUST** confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can’t be used! This **MUST NOT** be the case and execution of tests and Challenges **MUST** guarantee the quality, the completion and full usability by end users of the product!”

This ledger is the structural artefact that makes the above mandate auditable per-submodule, per-feature, per-platform, per-invariant.

Audit trail

Date	Author	Round	Schema	Notes
2026-05-18	Claude Opus 4.7	round 68	1.0.0	Initial publication. 68 rows (one per owned submodule). 8 PARTIAL Overall (AutoTemp, Storage, LLMOrchestrator×2, HelixSpecifier, LLMProvider, LLMsVerifier, Models — all backed by CONTINUATION rounds 23/37/50/63/64/65). 60 UNCONFIRMED Overall (honest baseline). Closes 50+ round deferred governance debt for CONST-048 ledger structure. Conservative population per CONST-035 anti-bluff posture.

Sources verified 2026-05-29: internal-governance doc — no third-party-service operator instructions to cross-reference. This ledger is HelixCode's own CONST-048 submodule × feature × platform × invariant tracker; it references internal constitutional anchors plus a few third-party identifiers used as captured-evidence records, not as instructions: opencode v1.14.41 (round-64 evidence of a real binary invocation — a historical evidence pin, deliberately not re-bumped), pgx/v5 / Redis / OpenTelemetry (stack-description notes). Per CONST-036, the LLM provider/model identifiers in these rows are LLMsVerifier-sourced at runtime (helixcode llm models list) — they are NOT pinned here and are deliberately left unmodified. Version authority per CLAUDE.md §3.1 (confirmed in-tree): inner module go 1.26, root go 1.25.2; PostgreSQL 15+; Redis 7+ — no stale build/Docker version pins in this doc to correct. Reviewed against the live tree on this date; no corrections required.